

BANARAS HINDU UNIVERSITY

Under Graduate Multidisciplinary Course Examination, 2024-25 (NEP)

COMPUTER SCIENCE**Paper Code: CSCMD11 : Fundamentals of Computing**

Time: 2:30 Hours

Full Marks: 70

*Instructions: Attempt any **five** questions. All questions carry equal marks (14 marks each). Write your Roll No. at the top immediately on receipt of this question paper.*

1. (a) Differentiate between computer hardware and software. What are the different types of computer software? Discuss in detail. Also give examples of each of them. [7]
(b) Define what is a computer. What are the different classifications of digital computer? Discuss. [7]
2. (a) What do you understand by an operating system? What is its role in the functioning of a computer? Also discuss the different types of Operating system in detail. [7]
(b) Differentiate between compiled and interpreted languages. Also give examples for each of them. [7]
3. (a) Discuss the different components of a network communication system. Also discuss the different types of data used for communication over network. [7]
(b) Discuss the different criteria used for judging the quality of a network. Also discuss the different types of networks in detail. Also give examples. [7]
4. (a) Discuss different types of network topologies. Also discuss their advantages and disadvantages. [7]
(b) What is Internet? Why is it important? Discuss the basic structure of Internet. [7]
5. (a) What do you understand by a data center? What is its role in the Internet? Explain with the help of examples. [7]
(b) What are Internet Service Providers (ISPs)? Also, discuss how last-mile connectivity is provided through the Internet. [7]
6. (a) Discuss how addressing is done on the Internet. Also differentiate between IPv4 and IPv6 addressing schemes. [7]
(b) Discuss the structure of World Wide Web in detail. [7]
7. (a) Differentiate between algorithm and flowchart. Also discuss the advantages and disadvantages of using a flowchart. [7]

(b) Draw a flowchart to print all even numbers between 1 and 20.

[7]